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Transmission-Mounted Power Control Unit Including 12-Volt DC-DC Converter for Two-Motor Hybrid System

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Abstract

This research proposes a third-generation power control unit (PCU) for a two-motor hybrid system. To make a more compact intelligent power unit (IPU) to be located under the second seat, a PCU with a 12-volt DC-DC converter (DCDC) that mounts directly on the transmission was developed, whereas the DCDC was previously mounted within the IPU. Since this has a considerable impact on the engine room layout, the technology described below was used to make the PCU even more compact than the second-generation unit. The power module, a key component of the PCU, now uses Ag nanoparticles sintering bonding rather than conventional solder bonding. This helps lower thermal resistance and enables smaller power semiconductors. The voltage control unit (VCU) has a new circuit that uses a multi-stage switching circuit and electric power transfer capacitor instead of the conventional chopper circuit. This makes it possible to shrink the reactor to less than 65% of its usual volume without raising the carrier frequency, and enables a layout that efficiently uses both sides of the water jacket (W/J), so that no specially designed W/J needs to be added just for the DCDC. As for the W/J seal, friction stir welding (FSW) to increase rigidity was used to reduce seal width and to make the W/J itself more compact. The use of these compactness technologies enabled the PCU to maintain all the efficiency of a second-generation PCU and made it possible to build the DCDC in the PCU with its volume less than a second-generation one. Development of this PCU makes the IPU much more compact and provides a similar amount of trunk space as in a gasoline-powered vehicle.

Introduction

Global warming caused by emissions from cars and air pollution can lead to serious issues, and each automotive industry is furthering the development of electric vehicles as the solution. The two-motor hybrid system that Honda developed in 2013 is proposed as one possible method of addressing these issues (Reference 1). The powertrain consists of an engine, an electric coupled continuously variable transmission (e-CVT), which has a built-in motor and generator, a power control unit (PCU), and an intelligent power unit (IPU), which consists of a high voltage battery (HV battery), an electric control unit (ECU) for HV battery control and an electric junction box. Furthermore, a second-generation two-motor hybrid system was proposed in 2016, and will be adapted for use in many kinds of electric vehicles, such as hybrid electric vehicles (HEV), plug-in HEVs (PHEV), electric vehicles (EV), and fuel cell vehicles (FCV), and a transmission-mounted PCU has been developed (Reference 2). In the second-generation system, a 12-volt DC-DC convertor (DCDC) is positioned inside the IPU. Because of the size restriction, the IPU is positioned in the trunk space, and therefore the cargo space is restricted compared with that of a gasoline-powered vehicle. Moreover, since it is necessary to have a cooling circuit that takes the DCDC into account, the degree of freedom of the IPU shape itself is also restricted, a problem that also impacts future capacity expansion of HV batteries for PHEV, etc. The new PCU will be adapted for use in many vehicle types, from sedans to sports utility vehicles, as the market for electric vehicles expands. A size reduction of the IPU is needed in order to maximize interior and cargo space. The author has developed a third-generation PCU for a two-motor hybrid system that is equipped with a DCDC that is smaller in volume than the second-generation PCU.

Overall Configuration and Basic Structure

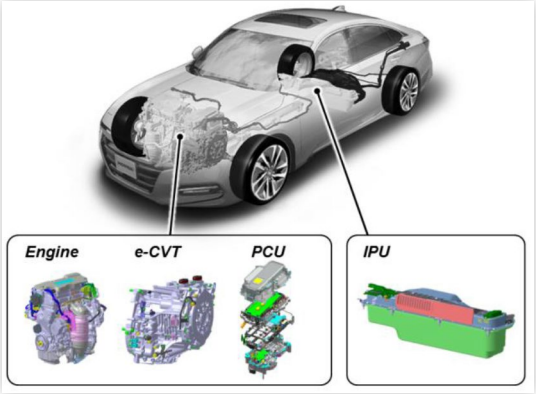
The layout of the third-generation PCU for a two-motor hybrid system is shown in Figure 1. The DCDC is moved from the IPU to the PCU and the miniaturized IPU is positioned under the second seat. The PCU is directly mounted on the transmission in the engine compartment, as with the second-generation PCU (Figure 2).

Block diagram of the third-generation PCU and structure of the PCU are shown in Figure 3 and Figure 4. As in the second-generation PCU, it consists of an electric control unit (ECU), which controls the inverter of the drive motor to drive

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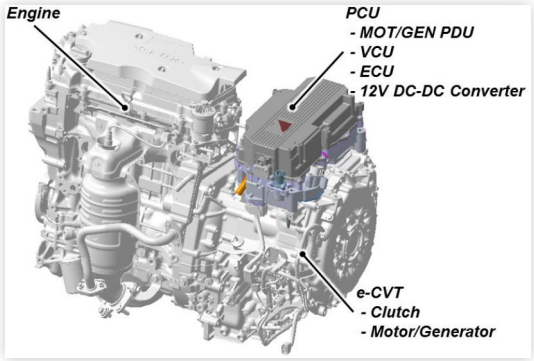
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FIGURE 1 Layout of PCU in two-motor hybrid system



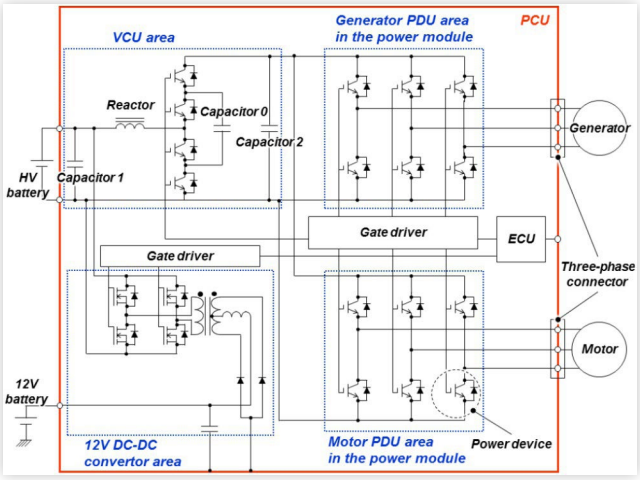
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FIGURE 2 Layout of hybrid power train



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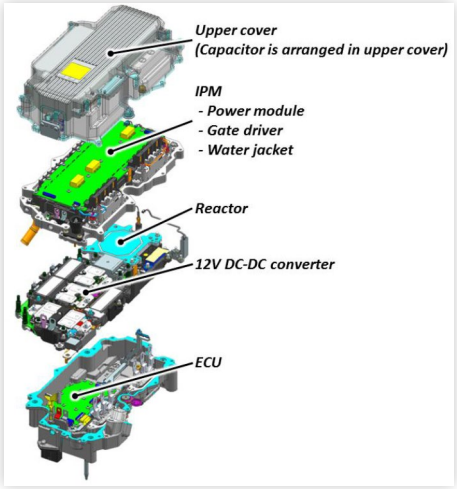
FIGURE 3 Block diagram of PCU



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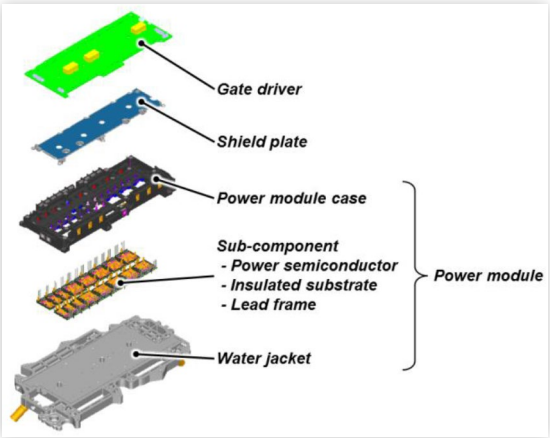
the vehicle and the generator for power generation, and a voltage control unit (VCU), which controls the system voltage optimally. An intelligent power module (IPM) comprises a power module, a gate driver and a water jacket (W/J), with a unified heat sink (H/S) and a middle case. Power semiconductors, which are used in the power dive unit (PDU) and VCU, are all in one package as a power module case (Figure 5).

FIGURE 4 Structure of PCU



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FIGURE 5 Structure of IPM



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Positioning the DCDC at the opposite side of the IPM enables both to share the W/J.

HEV System and PCU Specifications

The specifications for the second-generation and third-generation HEV systems and PCUs are shown in Table 1. In the third-generation PCU, although the DCDC is built in, miniaturization technologies used in the power module and VCU helped achieve an 8.8 L PCU capacity. This value is less than that of the second-generation system. Through this development, the size of the DCDC and its surrounding parts in IPU were reduced (Figure 6), and this contributes to a significant size reduction of the IPU (32% reduction), securing a cargo space equal to that of a gasoline-powered vehicle (Figure 7).

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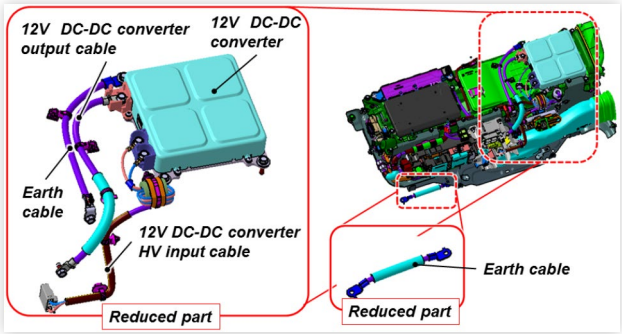
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TABLE 1 Specifications of HEV system and PCU

		Second generation	Third generation
HEV system	System maximum voltage [V]	700	650
	Motor maximum torque [Nm]	315	315
	Motor maximum output [kW]	135	135
	Generator maximum output[kW]	106.1	106.1
PCU	Maximum boost voltage [V]	700	650
	Motor maximum current [Arms]	320	310
	Generator maximum current [Arms]	155	155
	VCU maximum output[kW]	59.4	59.4
	12 V DC-DC converter maximum output [kW]	-	2.25
	Main components	Motor/ Generator inverter Gate driver ECU VCU	Motor/Generator inverter Gate driver ECU VCU 12 V DC-DC converter
	Weight [kg]	14.3	15.4
	Volume [L]	8.9	8.8

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FIGURE 6 Reduced parts in IPU

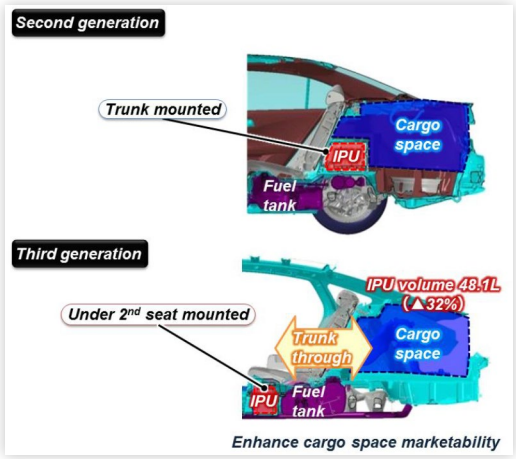


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Internal Component Optimization

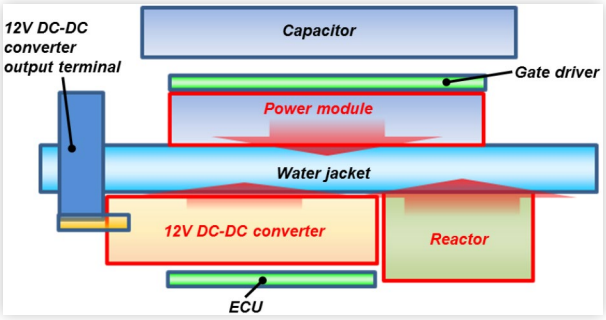
In order to build in the DCDC and have it mounted on the transmission case directly, heat resistance, vibration resistance and size reduction are issues that must be addressed. In the third-generation PCU, heat and vibration resistance were achieved by directly mounting a heat-generating and comparatively heavy power module, a reactor, and the DCDC on a highly rigid W/J sealed using FSW. Specifically, the height of the PCU was reduced by sandwiching the W/J between the power module and the reactor and DCDC, making it possible to reduce the size of the W/J. This allowed us to arrange a slim capacitor on the floor right above the power module (Figure 8). Moreover, the primary side capacitor of the DCDC was reduced in size by connecting the primary voltage line from the HV battery in the shortest way. These changes realized the development of a PCU that is smaller than that of the second-generation PCU, while the DCDC is built-in (Figure 9).

FIGURE 7 IPU downsizing effect



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FIGURE 8 Major component layout in PCU



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Power Module

The power module occupies a large portion of space in a PCU, and is one component that can make a significant contribution to size reduction. In order to maintain all of the efficiency of the second-generation PCU and to have the DCDC built-in, it is necessary to promote the reduction of the power module size, while maintaining its loss characteristic. As shown in Figure 5, the power module mainly consists of power semiconductors and a W/J. To reduce the size of the power module, we focused on reducing the power loss of the power semiconductors and the thermal resistance.

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FIGURE 9 High voltage DC connector layout

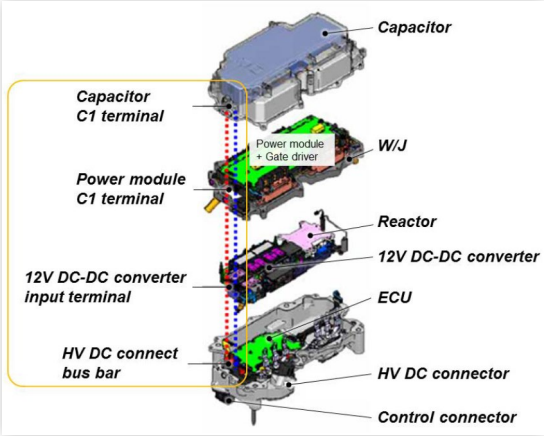
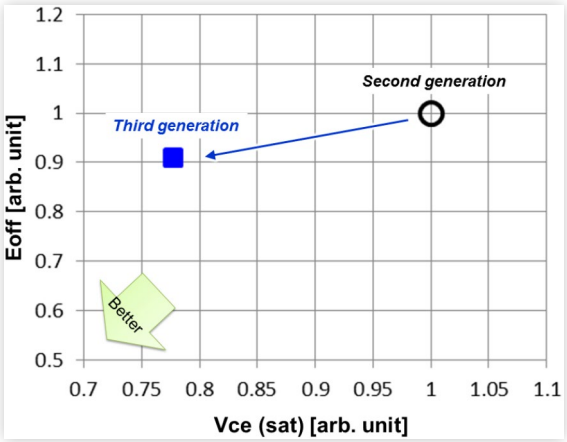


FIGURE 10 Normalized characteristics of IGBT



Reducing the Power Loss of the Power Semiconductors

As shown in Figure 10 and Figure 11, both the insulated gate bipolar transistor (IGBT) and the free-wheeling diode (FWD) chips used in the third-generation PCU have enhanced performance. In order to perform power semiconductor loss reduction, it is effective to make them wafer-thin. Regarding the IGBTs, the steady loss (Vce (sat)) and the turn-off switching loss (Eoff) have been reduced through ultra-thin wafer processing. Since thinning a wafer reduces the thermal capacitance and increases the saturated current, short-circuit protection becomes difficult because of lowered short-circuit tolerance. Therefore, the saturation current was suppressed and the generated energy at the time of a short circuit was reduced by optimization of the area ratio between the n + emitter layer and p + layer in the MOS Structure. This technology achieved extremely thin wafer formation and short circuit protection (Figure 12) (Reference 3). Regarding the FWDs, generation of the oscillation phenomenon during recovery action was controlled by adding a P layer/n + cathode layer formed partially on the cathode side using a new backside structure technology. The voltage surge of the FWDs during recovery states is

FIGURE 11 Normalized characteristics of FWD

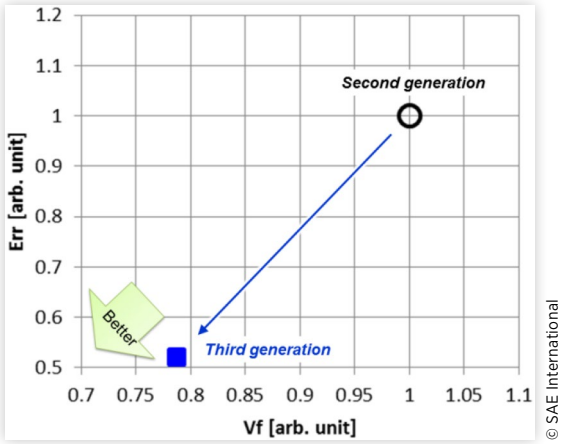


FIGURE 12 Structure of IGBT

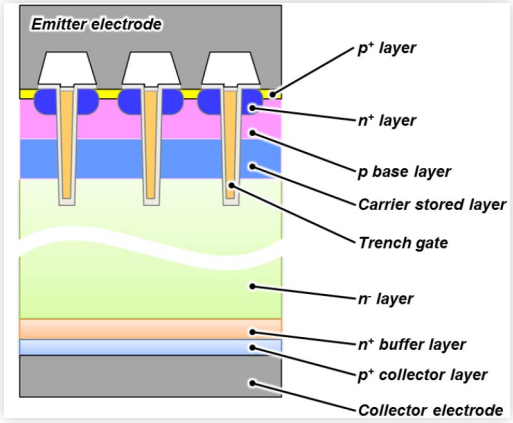
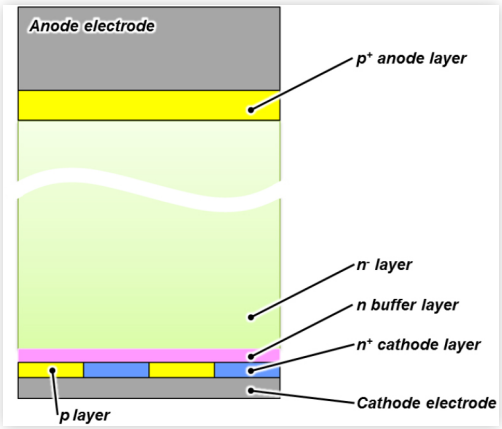


FIGURE 13 Structure of FWD

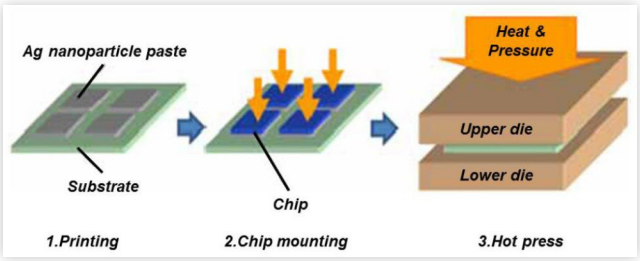


decreased through the use of this technology, and it becomes possible to reduce the silicon thickness, thus reducing the steady loss (Vf) and the reverse recovery loss (Err) (Figure 13) (Reference 3). The chip size was reduced by reducing the power loss of the power semiconductors while maintaining the loss characteristic of the power module through the application of these technologies.

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FIGURE 14 Process of Ag nanoparticle sintering bonding



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Reducing the Heat Resistance of Die Bonding

In the second-generation PCU, tin (Sn) solder was used to join (die bond) chips to the insulated substrate.

Regarding conventional solder bonding, fine voids gather at the time of solder smelting, larger voids are formed by aggregation, and these can have a harmful influence on the radiation performance. In the third-generation PCU, voids were reduced by using Ag (silver) nanoparticle sintering bonding for die bonding. In sintering bonding, the melting point depression phenomenon, in which the melting point lowers as the particle size becomes smaller, made it possible to join at a reflow temperature equivalent to that of the conventional soldering process.

The sintering bonding process is shown in Figure 14 (Reference 4). The bonding material, a paste-like material composed of Ag nanoparticles covered with an organic protective coating and several solvents, is used in this process. At first the paste-like material, which contains dispersed Ag nanoparticles, is screen-printed on the insulated substrate. When it is heated, the solvents vaporize and the chips are then mounted. Finally, the chips are hot-pressed to complete the sintering bonding. This hot press decomposes the organic protective coating, and the Ag nanoparticles come into contact and are sintered and bonded at a temperature that is lower than the melting point of Ag.

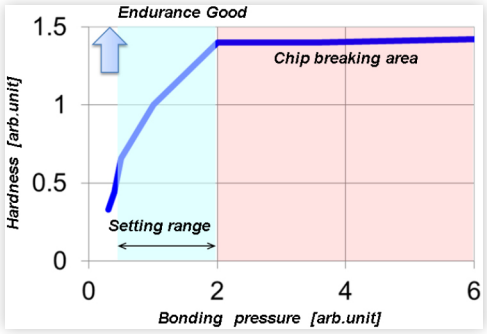
This process is based on a solid-state bonding unlike the conventional liquid-state bonding using solder, so large voids, which affect the thermal resistance, are not generated in the joints. Moreover, the joint thickness was reduced by 30% compared to that of conventional joints, because joint reliability is enhanced by using Ag, which is stronger than Sn, and the thickness is controlled by the bonding pressure.

The higher bonding pressure increases the joint strength and enhances the endurance reliability, but a bonding pressure that is too high can cause chips to break. An optimal bonding pressure was set (Figure 15). It was possible to reduce thermal resistance by 10% due to the reduction in voids and joint thickness, and the increased heat conductivity of Ag.

Cooling Structure

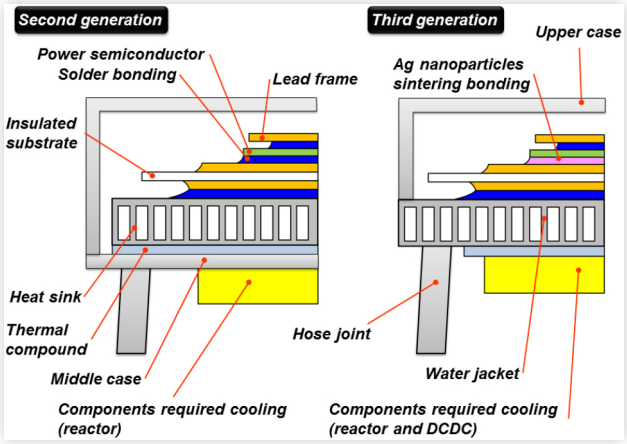
The cooling structure of the power semiconductors is the same as that of the second-generation PCU (H/S - solder - isolated substrate - die bonding - chip) (Figure 16). On the opposite side of the power module in the second-generation PCU, indirect cooling using a thermal compound was performed in the order H/S - thermal compound -middle case - reactor. In the third-generation PCU, because both the reactor and the DCDC release heat on opposite sides, a decrease in radiation performance through indirect cooling became an issue. By using a W/J that united the H/S and middle case, instead of a thermal compound, heat release performance was improved and the size of the DCDC was reduced. Since the seal, which includes a bolt connecting point, was wider in the conventional rubber seal method, FSW was adopted as a sealing method between the H/S and middle case. As a result, seal width was reduced by 10% (Figure 17). Moreover, the joint area at the center in the W/J enhanced W/J rigidity and endurance reliability for water pressure fluctuation and vibration.

FIGURE 15 Normalized sintering bonding hardness



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FIGURE 16 Intersection of cooling structure



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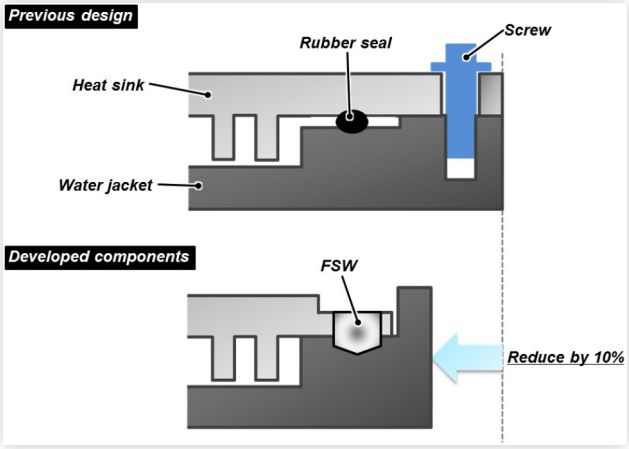
Voltage Control Unit (VCU)

Since the reactor and capacitors of the VCU also account for a large share of the PCU's internal volume, in order to reduce the size of the PCU, it is important to make these components smaller. In particular, since the heat release from the cooler is required for a reactor, it competes with the heat release of

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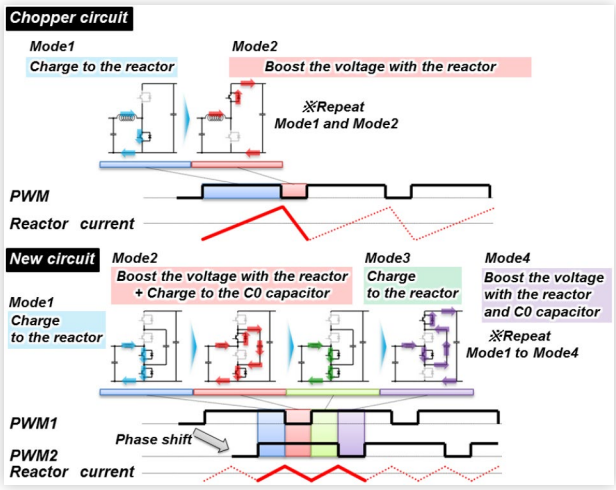
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FIGURE 17 Intersection of seal structure



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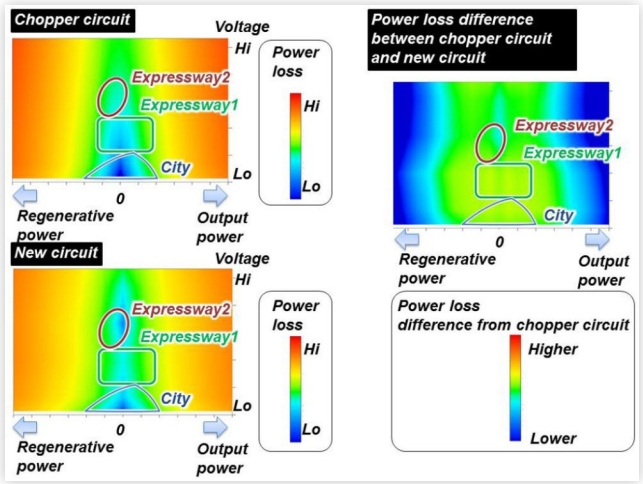
FIGURE 18 VCU switching pattern



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the DCDC. The DCDC is built into the PCU, so it is necessary to reduce the size of the reactor. Although it is common to increase the carrier frequency as a method of reducing the size of the reactor, the switching loss of power semiconductors increases and proper heat balance cannot be obtained. Therefore, in the third-generation PCU, rather than raise the carrier frequency, a new VCU circuit that uses a multi-stage switching circuit and capacitor for electric power transfer from a conventional chopper circuit was used to achieve a reactor size that is 65% or less that of a conventional one. It is a characteristic of this VCU circuit to have a united chopper circuit and capacitor shifting circuit, and to boost voltage using a reactor and C0 capacitor. The Hi and Lo sides have two arms respectively, and by shifting the phase of the PWM (pulse width modulation) signal by 180 degrees, the ripple current frequency that flows into the reactor double that of the carrier frequency, and the required inductance was halved. Furthermore, since the applied voltage for the reactor was reduced by the addition of C0, the inductance was reduced to 1/4 compared with that of a conventional circuit, and the reactor size was reduced by 65% (Figure 18).

FIGURE 19 VCU efficiency



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Another characteristic is that when the boost rate for secondary voltage is twice, the VCU power loss is low. The reason is as follows. Boost voltage by the reactor is unnecessary and the ripple current does not occur, so iron loss falls. Because the loss goes up as the boost rate goes up in a conventional chopper circuit, at high speeds, which require high boost voltage and a high MOT revolution range, VCU efficiency is low. By using this circuit, the loss is reduced in driving modes in which a high boost ratio is used by deciding on V2 voltage operation in consideration of this characteristic at twice the boost rate. Thanks to this new VCU circuit, downsizing of the reactor and efficiency improvement were realized (Figure 19).

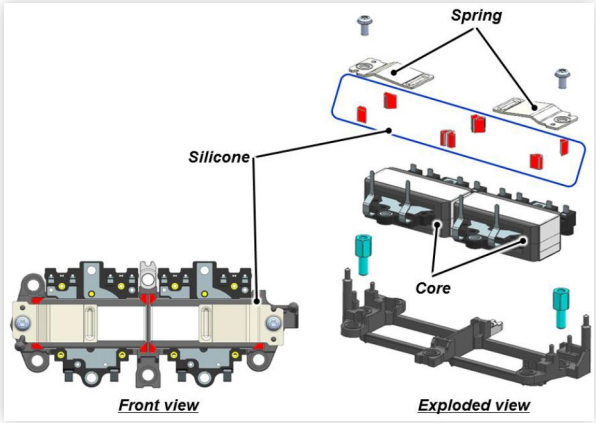
Transmission-Mounted Design

As with the second-generation PCU, the layout involves being mounted directly on the transmission and being connected to the motor through a three-phase connector (Reference 2). In order to build in the DCDC, heat and vibration transmitted through the transmission were dealt with. The heat was addressed by raising the cooling capability by adopting a water-cooling system for the DCDC instead of a conventional air-cooling system. As for vibration, input resonance with engine vibration dominant was avoided, and the PCU was designed to have a resonance characteristic on the higher-frequency wave side. The highly rigid W/J discussed in the Cooling Structure section is one of the methods used to raise the resonance point of the PCU. Moreover, the design concept for the substrate is also the same. Connecting points and the parts layout were designed so that the weak points of parts would not exceed the stress threshold level. The transformer and core of the smoothing coil, characteristic parts of DCDC, used magnetic components. A laminated spring holding structure, which is generally used for DCDC magnetic parts, was adopted. For transmission direct installation, silicone was poured at the four corners of the core (Figure 20). This was

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FIGURE 20 Countermeasures against vibration in the DCDC



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adopted for the purpose of controlling slight movements of the core caused by vibration and the generation of wear.

Using these applied technologies, a transmission-mounted PCU with a built-in DCDC was realized.

Summary

For the purpose of expanding cargo space, a third-generation PCU for two-motor hybrid systems was developed using the following technologies.

Heat release capability increase using Ag nanoparticle sintering bonding, and chip size reduction through chip loss reduction.

Reactor size reduction by adopting new VCU circuit.

Size reduction and rigid enhancement in W/J by adopting FSW.

Through the above miniaturization technologies, it was possible to build the DCDC into the PCU, resulting in a PCU smaller than the second-generation PCU, while the reduction of DCDC-related parts contributed to a 32% reduction in the size of the IPU compared with that of the second-generation IPU.

These technologies enhanced vehicle mount capacity and are applicable not only to HEV, but also to PHEV, EV, and FCV.

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